SDHMA

Std - X **Priliminary Examination** 2025 -26 **Marks - 40**

Time - 2.00 Hrs. **Sub : Science & Techcnology Part - 1** **Pages - 4**

Notes :- 1) All questions are compulsory.

2) Use of the calculator is not allowed.

3) The numbers to the right of the questions indicate full marks.

4) In case of MCQs (Q. No. 1 (A) only the 1st attempt will be evaluated and will be given credit.

5) For each MCQ the correct alternative (a), (b), (c) or (d) with subquestion number is to be written as an answer.
E.g. (i) (a), (ii) (b), (iii) (c)

6) Scientifically correct labelled diagrams should be drawn wherever necessary.

1. A) Answer write the correct alternative. **5**

1) ..... is one of the combustible compound of LPG.

a) Ethane b) Propane

c) Methane d) Ethene

2) The minimum velocity of spacecraft to escape from the earth's gravitational froce must be .....

a) 112 Km/s b) 11.2 Km/s

c) 1.12 Km/s d) 0.112 Km/s

3) Gold planted ornaments are example of .....

a) Electroplating b) Alloying

c) Anodizing d) Galvanizing

4) The ..... controls the ammount of light enentering the eye.

a) Iris b) Pupil

c) Cornea d) retina

5) .... light is deviated the maximum in the spectrum of white light obtained with a glass prisam.

a) Red b) Yellow

c) Violet d) Blue

https://www.maharashtrastudy.com

B) Answer the following questions. 5

1) State true or false.

In a period, the metallic characters increases from left to right

2) Write the co-relation between the two words.

IRNSS : Direction showing satellite : : INSAT : .....

3) Name the following.

Two isomers of butane

4) Find odd man out

Loud speaker, microphone, electric moter, magnet.

5) Match the correct one.

Column 'A'	Column 'B'
i) Refractive index of water	a) 1.31
	b) 1.36
	c) 1.33

2. A) Give scientific reasons. (Any 2) 4

1) A convex lens is called as converging lens.

2) Zero group elements are called nobel gases.

3) When silver vessels exposed to air turn blackish after some time.

B) Answer the following. (Any 3) 6

1) Write a note on free fall

2) Draw a neat lebelled diagrame of Hope's apparatus.

3) Differentiate between AC generator and DC generator

4) Calculate the ammount of heat required to convert 5 gm of ice of 0° C into water at 0° C.

5) Explain the reaction with the help of balanced equation.

i) Magnesium burns in air

ii) Calcium reacts with water

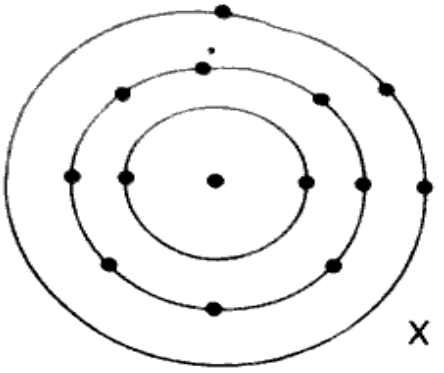
3. Answer the following questions. (Any 5) 15

1) Write the demerites of Mendeleev's periodic table

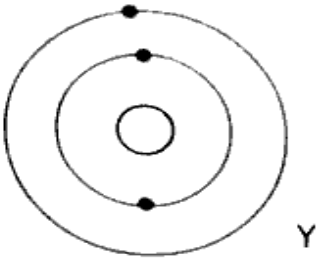
2) Explain the factors affecting the value of 'g'

https://www.maharashtrastudy.com

3)



X



Y

i) Identify the element 'X' and 'Y'

ii) Do these elements belong to same group.

iii) Which element is more electropositive in nature ? Why ?

4) Compleat the following table for convex lens.

Position of Object	Position of Image	Size of Image	Nture of Image
i)	At focus F_2	point image	Real and inverted
ii) At $2F_1$	At $2F_2$	.....	Real and inverted
iii) Between F_1 and O	on same side of object side	very large	.....

5) In std. X class out of 40 students, 10 students use spectacle. 2 student have positive power and 8 student have negative power of lenses in their spectacles.

Answer the following questions.

i) What does the negative power indicate?

ii) What defect of eye sight do most of the students suffer from?

ii) Generally which type of spectacles do most of the students use?

6) i) What is the principle behind the working of satellite lounch vehicle ?

ii) Write the formula for escape velocity

iii) Write the long form of ISRO

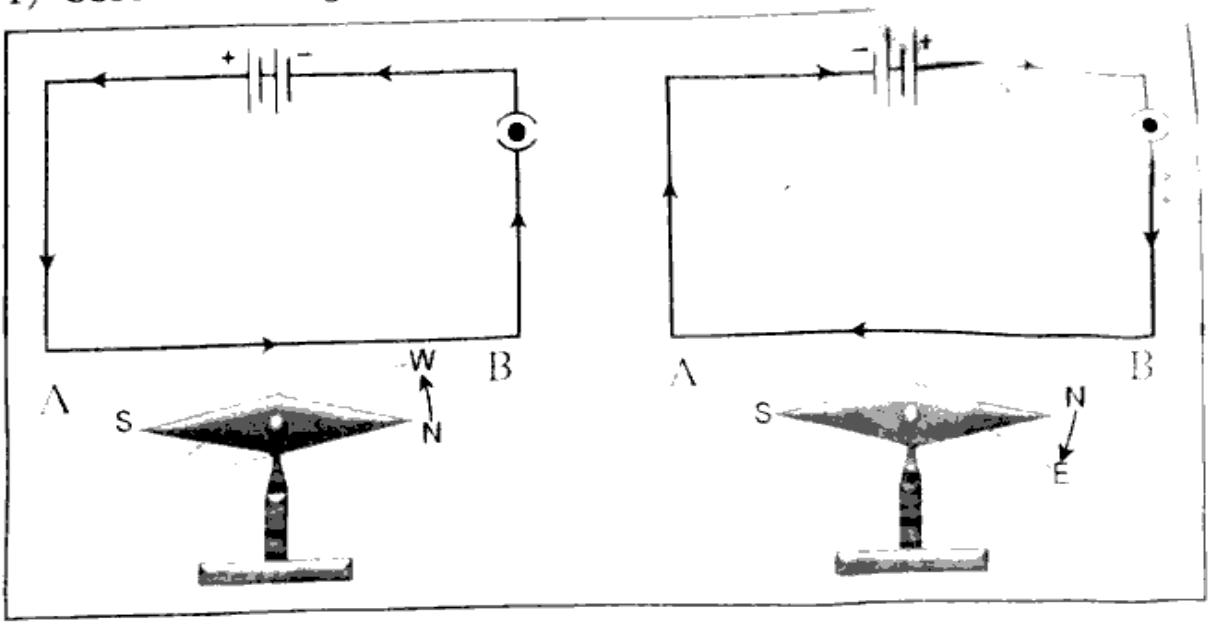
7) Complete the following table.

Common name	Structural formula	IUPAC name
i) ethylene	$CH_2 = CH_2$	.....
ii)	$CH_3 = COOH$	ethanoic acid
iii) methyl alcohol	.....	methanol

Lithiuam Lithium motion

4. Answer Any one of the following. 5

1) Observe the diagrams and answer the questio



i) Which effect of electric current is shown in the above figure. 1

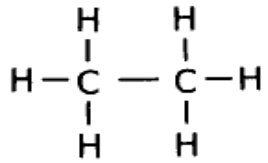
ii) What will happen if the number of electric cell is increased on the magnetic needle ? 1

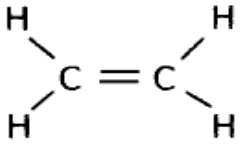
iii) If the distance between the conducter and magnetic needle is increased, what will be the effect on intensity of magnetic field ? 1

iv) If the ends of electric cell are interchanged what will be the effect on magnetic needle ? 1

v) Write the name of any two instruments which works on magnetic effect of electric current 1

2) i) Identify the saturated and unsaturated hydrocarbon from the given structural formula. 2

1) 

2) 

ii) Draw electron of structure for 1 and 2 2

iii) Define Homologous series. 1

https://www.maharashtrastudy.com